02 · Components Buying List (BOM)

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> **Prices are estimates** at a 100-unit run (with about 10 % overage for SMT attrition and test spares) and were not live-quoted in this session. Before ordering, confirm every line with a PCBWay turnkey quote or a DigiKey/LCSC cart. Links go to the vendor search for the exact manufacturer part number, so they do not break when a product page moves.

EVT / PVT BUILD (6.5 MM, PRE-CERTIFIED MODULE)

Ref	Part	MPN	Qty/unit	Order (100 run)	Est. $/unit	Est. line $	Buy
U1	ESP32-S3 module 4MB flash 2MB PSRAM (EVT)	`ESP32-S3-MINI-1-N4R2`	1	110	$3.20	$352.00	LCSC/DigiKey
DS1	Sharp Memory LCD 1.08in 160x68	`LS011B7DH03`	1	105	$9.50	$997.50	Mouser/DigiKey
J1	0.5mm FPC connector 10-pin back-flip	`Hirose FH34SRJ-10S-0.5SH`	1	110	$0.60	$66.00	DigiKey
CAM1	Ultra-thin VGA camera module DVP FPC (<=2.5mm Z)	`GC032A / GC0308 FPC module`	1	110	$3.00	$330.00	LCSC/Alibaba module houses
J2	0.5mm FPC connector 24-pin	`Hirose FH34SRJ-24S-0.5SH`	1	110	$0.80	$88.00	DigiKey
U2	Environmental sensor T/RH/P/gas with AI-Studio	`Bosch BME688`	1	110	$7.80	$858.00	DigiKey/Mouser
U3	6-axis IMU with on-chip gesture/wrist-wake AI	`Bosch BMI270`	1	110	$2.60	$286.00	DigiKey
U4	Ambient light sensor	`Lite-On LTR-303ALS-01`	1	110	$0.70	$77.00	DigiKey
U5	Qi receiver + Li-ion charger single chip	`TI BQ51050BRHLR`	1	110	$3.10	$341.00	DigiKey/TI
L1	Qi receiver coil ~20x20mm thin with ferrite	`TDK WR202010-18M8-G (or Wurth WE-WPCC equiv)`	1	110	$2.50	$275.00	DigiKey
U6	Ultra-low-Iq buck 3.3V 750mA	`TI TPS62840DLCR`	1	110	$0.95	$104.50	DigiKey
U7	Fuel gauge ModelGauge	`Analog MAX17048G+T10`	1	110	$1.40	$154.00	DigiKey
U8	Battery protection IC	`ABLIC S-82D1AAA-A8T2U7 (or DW01 equiv)`	1	110	$0.35	$38.50	LCSC
BT1	LiPo cell EVT 3.0x30x30mm ~150mAh with PCM	`303030 class`	1	110	$2.20	$242.00	Alibaba (Grepow/PKCELL/EEMB)

Ref	Part	MPN	Qty/unit	Order (100 run)	Est. $/unit	Est. line $	Buy
U9	Haptic driver (LRA/ERM licensed library)	TI DRV2605LDGSR	1	110	$1.60	$176.00	DigiKey
M1	Z-axis LRA 8mm dia x 2.5mm	Jinlong G0825001 / Vybronics VG0825001	1	110	$1.80	$198.00	DigiKey
SW1	Low-profile tact switch (Copy)	C&K KMR211NG LFS (or Alps SKRPABE010)	1	110	$0.45	$49.50	DigiKey
J3	Tag-Connect TC2030 footprint (no part)	TC2030-IDC-NL footprint	1	0	$0.00	$0.00	Tag-Connect
PASS	Caps/resistors/ferrites 0201-0402 kit	various	1	110	$0.80	$88.00	PCBWay turnkey
PCB	Main PCB 4-layer 0.6mm 36x36mm HDI ENIG	custom	1	110	$4.00	$440.00	PCBWay
FPC	Flex for camera/antenna/LRA (optional)	custom	1	110	$1.50	$165.00	PCBWay
ENC-A	Back tray 316L CNC + #8 mirror polish + laser LOT mark	custom	1	110	$18.00	$1,980.00	PCBWay CNC
ENC-B	Front bezel 316L CNC + bead-blast/brush	custom	1	110	$12.00	$1,320.00	PCBWay CNC
GL1	Front cover glass 0.4mm chemically strengthened + black mask print	custom	1	110	$2.50	$275.00	Alibaba cover-glass vendors
GL2	Back Qi window 26mm dia 0.5mm glass + NCVM mirror-silver	custom	1	110	$3.00	$330.00	NCVM cover-glass vendors
VENT	ePTFE adhesive vent membrane 3mm	Gore PolyVent / Donaldson Tetratex equiv	1	110	$0.60	$66.00	DigiKey
GSK	Die-cut gasket + PSA (3M 300LSE / VHB)	custom die-cut	1	110	$0.50	$55.00	PCBWay / local die-cut
SCR	M1.0 x 2.5 Torx T3 stainless screws	custom/standard	4	450	$0.06	$27.00	McMaster/Alibaba
KEY	Steel Copy key cap + silicone plunger	custom	1	110	$1.20	$132.00	PCBWay CNC + silicone vendor
CHG	Qi 5W charging pad (EVT off-the-shelf)	any WPC-certified 5W pad	1	105	$4.00	$420.00	Alibaba/Amazon
CBL	USB-C cable 1m braided	generic	1	105	$1.20	$126.00	Alibaba
PKG	Box + insert + printed quick-start card	custom	1	105	$3.50	$367.50	PCBWay / print vendor

Estimated material cost per unit: $95.59 · Estimated material for the 100-unit run: $10,424.50

Not included above: PCBA labour and stencil (07 §3), CNC set-up/NRE, fixtures, certification and freight. See the full run budget in 08 · Roadmap §5.

DVT SUBSTITUTIONS FOR THE 5.0 MM TARGET

Ref	Part	MPN	Est. $/unit	Buy	Notes
U1-DVT	ESP32-S3 SoC QFN56 4MB flash 2MB PSRAM (DVT 5mm)	ESP32-S3FH4R2	$2.40	LCSC	needs 40MHz XTAL + chip antenna + own FCC grant
Y1	40MHz crystal 2016 (DVT only)	TXC 8Y40000013	$0.15	LCSC	DVT only
BT1-DVT	Custom ultra-thin LiPo 1.6x30x30mm ~70mAh	custom (Grepow / EEMB)	$4.50	Grepow	custom MOQ often 500-1000; quote at 100 with NRE
ANT1	2.4GHz chip antenna (DVT only)	Johanson 2450AT18B100	$0.40	DigiKey	EVT module has PCB antenna -> needs non-metal window too

ORDERING SEQUENCE

1. **Long lead and custom first (week 0):** cover glass (GL1), NCVM window (GL2), LiPo cell (BT1, needs UN38.3 docs), camera module (CAM1, request a Z-height drawing from 3 vendors).
2. **Dev kits now (week 0, about $150):** ESP32-S3-DevKitC-1-N8R8, Adafruit Sharp Memory display breakout (LS011B7DH03-based, product 3502 class), Pimoroni/Adafruit BME688 breakout, Adafruit DRV2605L breakout with LRA, TI BQ51050BEVM-764 Qi receiver EVM, any Qi 5 W pad.
3. **PCBA turnkey (week 4):** upload Gerbers, the centroid file and this BOM to PCBWay. Let PCBWay source the passives and commodity ICs. Consign the display, camera and cell.
4. **CNC (week 4, in parallel):** 5 sample enclosures first (finish check), then 110.

DEV-KIT BREADBOARD LIST (EVT-0, NO CUSTOM PCB)

Item	Why	Link
ESP32-S3-DevKitC-1-N8R8	MCU, Wi-Fi, PSRAM, camera pins	https://www.digikey.com/en/products/result?keywords=ESP32-S3-DEVKITC-1-N8R8
Sharp Memory LCD breakout	Screen and driver code	https://www.adafruit.com/search?q=sharp+memory+display
BME688 breakout	Weather and AI gas	https://www.digikey.com/en/products/result?keywords=BME688+breakout
DRV2605L haptic breakout and LRA	Pager buzz	https://www.adafruit.com/search?q=DRV2605L
BMI270 breakout	Pickup-to-wake	https://www.digikey.com/en/products/result?keywords=BMI270+shuttle
OV2640/GC032A camera on FPC	Capture path	https://www.lcsc.com/search?q=GC032A
TI BQ51050BEVM-764	Qi receive and charge	https://www.ti.com/tool/BQ51050BEVM-764
150 mAh 303030 LiPo	Battery	https://www.adafruit.com/search?q=lipo+150mAh